

ME404: Dynamics and Control of Mechanical Systems, Fall 2026

Problem Set #1: Modeling and Dynamics

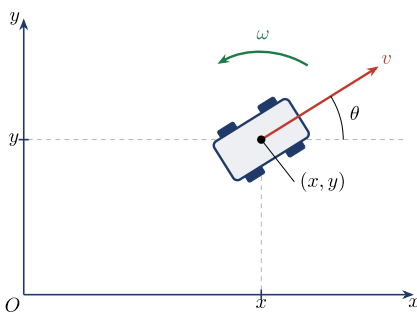
Due Date: 3:30 PM, Tuesday, September 29, 2026

Problem #1

(Part A) Dubin's Kinematics (NO AI)

A wheeled cart moves in a horizontal plane. Let (x, y) be the position of the cart's center in a fixed inertial frame, and let θ be the vehicle heading, measured counterclockwise from the x -axis. The cart is driven forward along its heading with speed $v(t)$ and turns with angular rate $\omega(t)$. Assume the wheels roll without slipping, so the cart's velocity is always directed along its heading — it cannot translate sideways.

Taking the state to be $q = (x, y, \theta)$ and the inputs to be $u = (v, \omega)$, derive the differential equations $\dot{q} = f(q, u)$ that describe the kinematics of the cart.



(Part B) Dubin's Dynamics (NO AI)

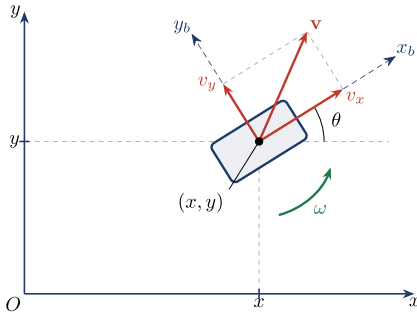
The vehicle of Part A is now free to slip sideways — model it as a rigid body of mass m moving in a horizontal plane, with moment of inertia J about the vertical axis through its center of mass.

Attach a body frame at the center of mass: the x_b -axis points forward along the heading, the y_b -axis points to the vehicle's left, and the z_b -axis points out of the plane, forming a right-handed triad. As before, θ is the heading measured counterclockwise from the inertial x -axis, and $\omega = \dot{\theta}$. Let (v_x, v_y) be the vehicle's velocity **expressed in the body frame**.

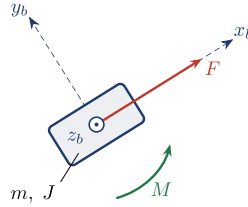
The vehicle is driven by a force $F(t)$ applied at the center of mass along the body x_b -axis, and a moment $M(t)$ about the body z_b -axis. Neglect friction and aerodynamic drag.

(a) Apply Newton's and Euler's laws to derive differential equations for v_x , v_y , and ω . Be careful: the body frame *rotates*, so the inertial acceleration is not simply the time derivative of the body-frame velocity components.

(b) Combine your result with the kinematics of Part A to write a full state model $\dot{q} = f(q, u)$, with state $q = (x, y, \theta, v_x, v_y, \omega)$ and input $u = (F, M)$.



(a) velocity in the body frame



(b) applied force and moment

Problem #2 (NO AI)

A two-link planar arm moves in a **vertical** plane under gravity g , as shown. Link 1 is pinned to a fixed base at O ; link 2 is pinned to the far end of link 1.

Let θ_1 be the angle of link 1, measured counterclockwise from the horizontal, and let θ_2 be the angle of link 2 **measured counterclockwise from link 1**, so that $\theta_2 = 0$ means the two links are collinear.

Link i ($i = 1, 2$) has length l_i and mass m_i , and its center of mass lies a distance l_{ci} from the joint at its base. Let J_i be the moment of inertia of link i about its own center of mass. A motor at each joint applies torque τ_1 at the base and τ_2 at the elbow. Neglect joint friction.

(a) Write the positions of both centers of mass in terms of θ_1 and θ_2 , differentiate to obtain their velocities, and form the total kinetic energy T and potential energy V of the arm.

(b) Using Lagrange's equations, derive the nonlinear equations of motion. Arrange them in the standard form

$$M(\theta) \ddot{\theta} + C(\theta, \dot{\theta}) \dot{\theta} + g(\theta) = \tau,$$

where $\theta = (\theta_1, \theta_2)^T$ and $\tau = (\tau_1, \tau_2)^T$, and identify the inertia matrix $M(\theta)$, the velocity-coupling matrix $C(\theta, \dot{\theta})$, and the gravity vector $g(\theta)$.

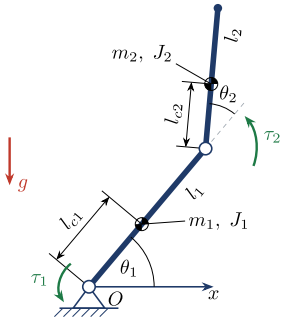
(c) Identify which terms in your result are responsible for the nonlinearity, and explain physically what the off-diagonal entries of $M(\theta)$ represent. Verify that $M(\theta)$ is symmetric.

(d) Verify that $\theta_1 = \pi/2$, $\theta_2 = 0$ — both links pointing vertically upward — is an equilibrium of the system with $\tau = 0$. Introducing the deviation coordinates $\varphi_1 = \theta_1 - \pi/2$ and $\varphi_2 = \theta_2$, linearize your equations of motion about this equilibrium to obtain

$$M_0 \ddot{\varphi} + K \varphi = \tau,$$

and give M_0 and K explicitly in terms of the link parameters. What becomes of the $C(\theta, \dot{\theta}) \dot{\theta}$ term, and why?

(e) Use your linearized model from (d) and determine the *four* transfer functions (one for each pair of inputs and outputs). What is the characteristic polynomial?



Problem #3 (AI ONLY ALLOWED ON PARTS E and F)

The system shown below consists of two carts of mass m_1 and m_2 rolling without friction on a horizontal surface; the rollers are massless. Cart 1 is connected to the left wall by a spring k_1 , cart 2 to the right wall by a spring k_3 , and the carts are connected to each other by a spring k_2 in parallel with a viscous damper b . A force $u(t)$ acts on cart 1 in the positive direction.

Let x_1 and x_2 be the cart displacements, measured positive to the right from the configuration in which all three springs are at their natural lengths. The springs are linear, the damper is linear viscous, and the damper is the only source of dissipation.

(a) Draw a free-body diagram for each cart and derive the two coupled equations of motion. Explain why b and k_2 appear in *both* equations, and why they appear only through the difference $x_1 - x_2$.

(b) Take Laplace transforms of both equations with zero initial conditions, and solve the resulting pair of algebraic equations for $X_1(s)$ and $X_2(s)$. Report the two transfer functions $X_1(s)/U(s)$ and $X_2(s)/U(s)$, and the characteristic polynomial $\Delta(s)$, in terms of the physical parameters. Identify the zeros of each transfer function and explain physically why X_2/U has only one zero while X_1/U has two.

(c) Take $m_1 = m_2 = 1$ kg, $k_1 = k_3 = 10$ N/m, $k_2 = 5$ N/m, $b = 0.5$ N·s/m. Show that $\Delta(s)$ factors into two quadratics, and find the natural frequency and damping ratio of each. What is unusual about one of them?

(d) Now let $u \equiv 0$ and allow nonzero initial displacements $x_1(0) = a_1$, $x_2(0) = a_2$, with both initial velocities zero. Re-transform the equations of motion, *retaining the initial-condition terms*, and solve for $X_1(s)$ and $X_2(s)$. Using the parameters of part (c), evaluate your expressions for each of these three cases and simplify completely:

1. $a_1 = a_2 = 1$ m
2. $a_1 = 1$ m, $a_2 = -1$ m
3. $a_1 = 1$ m, $a_2 = 0$

Something notable happens in cases 1 and 2. What is it, and what does it tell you about which modes each initial condition excites?

(e) (AI OKAY) Invert your transforms from part (d) to obtain $x_1(t)$ and $x_2(t)$ in closed form for all three cases, and plot them over $0 \leq t \leq 30$ s. Account physically for the behavior in case 1 — what is the damper doing? Verify that case 3 is the superposition you would expect from cases 1 and 2.

(f) (AI OKAY) Repeat cases 1 and 3 for $b = 0$, $b = 2$, and $b = 10$ N·s/m, and then again with m_2 changed to 3 kg (keeping $b = 0.5$). Describe how each response changes. Why does breaking the

symmetry overturn the conclusion you reached in part (e)?

(g) With $m_2 = 3$ kg and the other parameters as in part (c), use the final value theorem on $X_1(s)/U(s)$ and $X_2(s)/U(s)$ to predict the steady-state displacements under a unit step force. Verify both predictions by a static force balance on the two carts, and confirm them against a simulated step response.

